

ADSP 31014 Statistical Models for Data Science
Homework 1

Let's consider two discrete random variables X and Y with the following (joint) probability mass function.

x	y	$p_{X,Y}(x,y) = P(X = x, Y = y)$
-1	1	0.06
-1	2	0.10
-1	4	0.04
0	1	0.21
0	2	0.35
0	4	0.14
2	1	0.03
2	2	0.05
2	4	0.02

(5 points) Question 1

Compute the (marginal) probability mass function of X by “summing over the probabilities of Y ”.

$$p_X(x) = P(X = x) = \sum_{\text{all } y_j \text{ values}} P(X = x, Y = y_j)$$

x	$p_X(x) = P(X = x)$
-1	?
0	?
2	?

(5 points) Question 2

Compute the expectation $\mathbb{E}(X)$ of X .

$$\mu_X = \mathbb{E}(X) = \sum_{\text{all } x_i \text{ values}} x_i P(X = x_i)$$

(5 points) Question 3

Compute the variance $\text{Var}(X)$ of X .

$$\text{Var}(X) = \mathbb{E}(X - \mu_X)^2 = \sum_{\text{all } x_i \text{ values}} (x_i - \mu_X)^2 P(X = x_i)$$

(5 points) Question 4

Compute the standard deviation $\text{SD}(X) = \sqrt{\text{Var}(X)}$ of X .

(20 points) Question 5

Similarly, compute the (marginal) probability mass function $p_Y(y)$, expectation $\mathbb{E}(Y)$, variance $\text{Var}(Y)$, and standard deviation $\text{SD}(Y)$ of Y .

(5 points) Question 6

Compute the covariance $\text{Cov}(X, Y)$ of X and Y .

$$\text{Cov}(X, Y) = \mathbb{E}((X - \mu_X)(Y - \mu_Y)) = \sum_{\text{all } x_i \text{ and } y_j \text{ values}} (x_i - \mu_X)(y_j - \mu_Y) P(X = x_i, Y = y_j)$$

(5 points) Question 7

Compute the correlation $\text{Corr}(X, Y)$ of X and Y .

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\text{SD}(X) \text{SD}(Y)}$$

(5 points) Question 8

Based on Question 7, are random variables X and Y correlated or uncorrelated?

(5 points) Question 9

Based on Question 1 and 5, if we write the values of the marginal probability mass functions of X and Y as column and row vectors respectively

$$\vec{p}_X = \begin{pmatrix} ? \\ ? \\ ? \end{pmatrix} \quad \vec{p}_Y = (? \quad ? \quad ?)$$

Perform matrix multiplication $\vec{p}_X \times \vec{p}_Y$ to obtain a 3×3 matrix.

(5 points) Question 10

If we write the values of the joint probability mass function of (X, Y) as a 3×3 matrix

$$p_{X,Y} = \begin{pmatrix} 0.06 & 0.10 & 0.04 \\ 0.21 & 0.35 & 0.14 \\ 0.03 & 0.05 & 0.02 \end{pmatrix}$$

What is the relationship between this matrix and the one in Question 9? (Hint: The answer should be simple and obvious.)

(5 points) Question 11

Based on Question 10, are random variables X and Y independent or not?

(5 points) Question 12

To model the relationship of positive quantities (e.g. x = “infection rate of an apocalyptic virus” and y = “number of zombies”), we sometimes consider an exponential growth model form

$$y = \beta_0(x^{\beta_1})\epsilon$$

with parameters $\beta_0 > 0$ and $\beta_1 \in \mathbb{R}$, and a multiplicative positive noise term $\epsilon > 0$. Can this model be viewed as a linear model? Explain. (Hint: Take logarithm.)

(5 points) Question 13

Write a GenAI prompt to describe our Chicago taxi trip data set and its nature (e.g. column names) and ask some GenAI tool (e.g. ChatGPT, Gemini, Claude) for potential data issues to be aware of when performing data cleaning and exploratory data analysis on this data set. What does GenAI say? How good are the suggestions? Is there any suggestion that we have not thought of?